Project plan:
Super-resolution sea ice concentration using generative AI

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Project purpose:

The enhancement of SIC resolution is critical for improving short-term sea ice forecasting, navigation, and climate monitoring in the rapidly evolving Arctic environment. Traditional passive microwave (PMW) remote sensing is limited by coarse spatial resolution obscuring fine-scale features in sea ice maps like leads and floe boundaries essential for safe maritime operations. This project addresses these limitations by leveraging recent advancements in deep generative modeling to perform a dual-objective task: non-linear geophysical retrieval of Sea Ice Concentrations (SIC) and spatial super-resolution.

This project will be a feasibility study of the robustness of applying advanced generative models to remote sensing of sea ice and pave the way towards a 5 km Climate Data Record (CDR) exploiting the upcoming Copernicus Imaging Microwave Radiometer (CIMR) mission. By developing a framework that links legacy and future satellite capabilities, this research ensure continuity and enhanced precision in monitoring the Arctic's response to global warming.

Aims / Objectives:

The primary aim of this project is to develop a generative framework that accepts coarse-resolution PMW Brightness Temperatures (BT) as input and produces 5 km super-resolution sea ice concentration fields as output. This will demonstrate the feasibility of scaling deep generative models to enhance the spatial resolution of sea ice products.

To achieve this, the project will pursue the following key objectives:

- Establishing a strong theoretical and technical foundation through a review of existing literature and developing a thorough data familiarity with the AMSR2 and DMI-ASIP dataset.
- Design and implement a deep learning architecture capable of non-linear geophysical retrieval and spatial super-resolution.
- Optimize the model through iterations of training and fine-tuning to ensure the model's handling of seasonal variation and sub-pixel ice features.
- Evaluate the model's performance and robustness using statistical metrics (e.g. RMSE) and meteorological validation to ensure the generated maps are physically consistent with observed Arctic conditions.

Applied techniques:

The skills required to execute this project are within deep learning and remote sensing, including end-to-end development, training and optimization of neural networks and the retrieval of sea ice concentration from passive microwave brightness temperatures and the application of super-resolution techniques to enhance the spatial output.

To ensure rigorous evaluation and prevent data leakage, the dataset will be partitioned temporally. A training set of 1-2 years (e.g. 2020-2021) will be utilized for model development and fine-tuning, while a dedicated one-year test set (e.g. 2022) will be reserved for final performance benchmarking. By training the framework on a period characterized by extreme melt and record-low maximums, the model's robustness and its ability to generalize across modern climate anomalies and thinning ice regimes can be verified.

The SIC network will be U-net inspired with an encoder-decoder architecture, but the archival of super-resolution is not determined as multiple solutions have been identified during the literature review.

Materials:

The dataset for this project consists of temporal and spatially co-located Brightness Temperatures (BT) and high-resolution ice charts spanning the decade from 2014 to 2024. The BT are derived from the Advanced Microwave Scanning Radiometer 2 (AMSR2) and cover the spectral bands from 6.9 to 36.5 and 89.0 GHz, resampled to a 2 km spatial grid, which will be the input to the model. The high-resolution sea ice maps produced by Danish Meteorological Institute's Automated Sea Ice Products (DMI-ASIP) represented on a 80 m grid will serve as the ground truth for model training and evaluation.

Furthermore DMI will provide the necessary backend infrastructure and server capacity as part of the joint technical resource allocation.

Analysis of data:

The analysis for this project follows a dual-objective approach with achieving spatial super-resolution and retrieving of SIC. The framework will be designed to bridge the gab between the 2 km AMSR2 BT input and the 80 m DMI-ASIP SIC labels, by performing a non-linear inversion of multi-spectral signatures (6.9-89.0 GHz) to retrieve SIC and reconstructing high-frequency spatial features, such as leads and floe boundaries, that are not explicitly defined in the coarse input data.

Implementing this analysis requires a specialized stack, utilizing PyTorch for model development alongside other Python tools (e.g. Xarray and GDAL) for precise spatial alignment of the satellite data. The analysis will focus on a 1-2 year sub-dataset to balance computational efficiency with representative climate conditions.

Due to the density of the data regarding the 80 m grid resolution of the sea ice maps, the project will leverage DMI's high-performance computing infrastructure, which allows for GPU-accelerated training and fine-tuning, ensuring that the model achieves geophysical robustness and meets the technical milestones within the time frame.

Project risks:

Due to the time available, a full CDR will not be within reach, whereas the initial goal is limited to a proof-of-concept for one year to validate the models handling of the seasonal variation and to mitigate schedule risk to ensure on-time completion.

A part from the time aspect the primary project risk centers on the training and fine-tuning of the generative framework, where any shortfall in model performance may result in outputs not meeting the success criteria.

Report writing:

Part of the report writing will be simultaneously with implementing the model, as the training phase leave room for other tasks to be carried out.

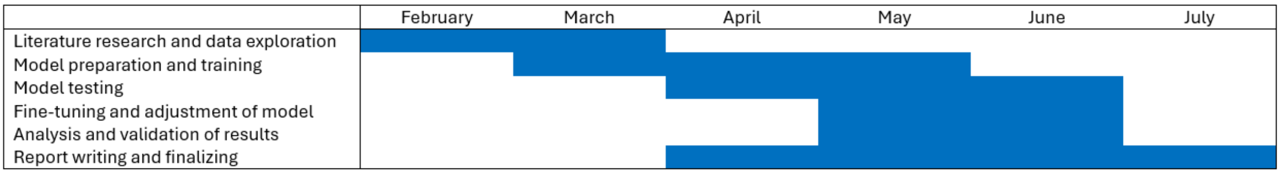
As the literature review and data exploration is almost done, these elements in the report will be drafted shortly hereafter.

Innovation/ IPR / Collaboration agreements:

This project is conducted together with The National Center for Climate Research (NCKF) of DMI using data belonging to them, hence the rights to the outcomes of this project is also reserve to them. This project is not considered to be relevant to patent.

Time plan:

The preliminary plan for the project is seen below. The plan is to have results by May in order to have time to adjust and fine-tune the model, before validating the final results.



Literature:

Dataset: `https://data.marine.copernicus.eu/product/SEAICE_ARC_PHY_AUTO_L3_MYNRT_011_023/description`
DMI-ASIP: `https://doi.org/10.1016/j.rse.2026.115252`
Network architecture: `https://doi.org/10.1109/TGRS.2020.3004539` and `https://doi.org/10.1109/JSTARS.2022.3232533`